# 중간고사 결과
# & 후보물질 탐색 및 발굴2

Han-Jun Kim (김한준), D.V.M., Ph.D.

Associate Professor
College of Pharmacy, Korea University

Email: hanjun@korea.ac.kr
Web: hanjunkim.co.kr

2026. 11. 04

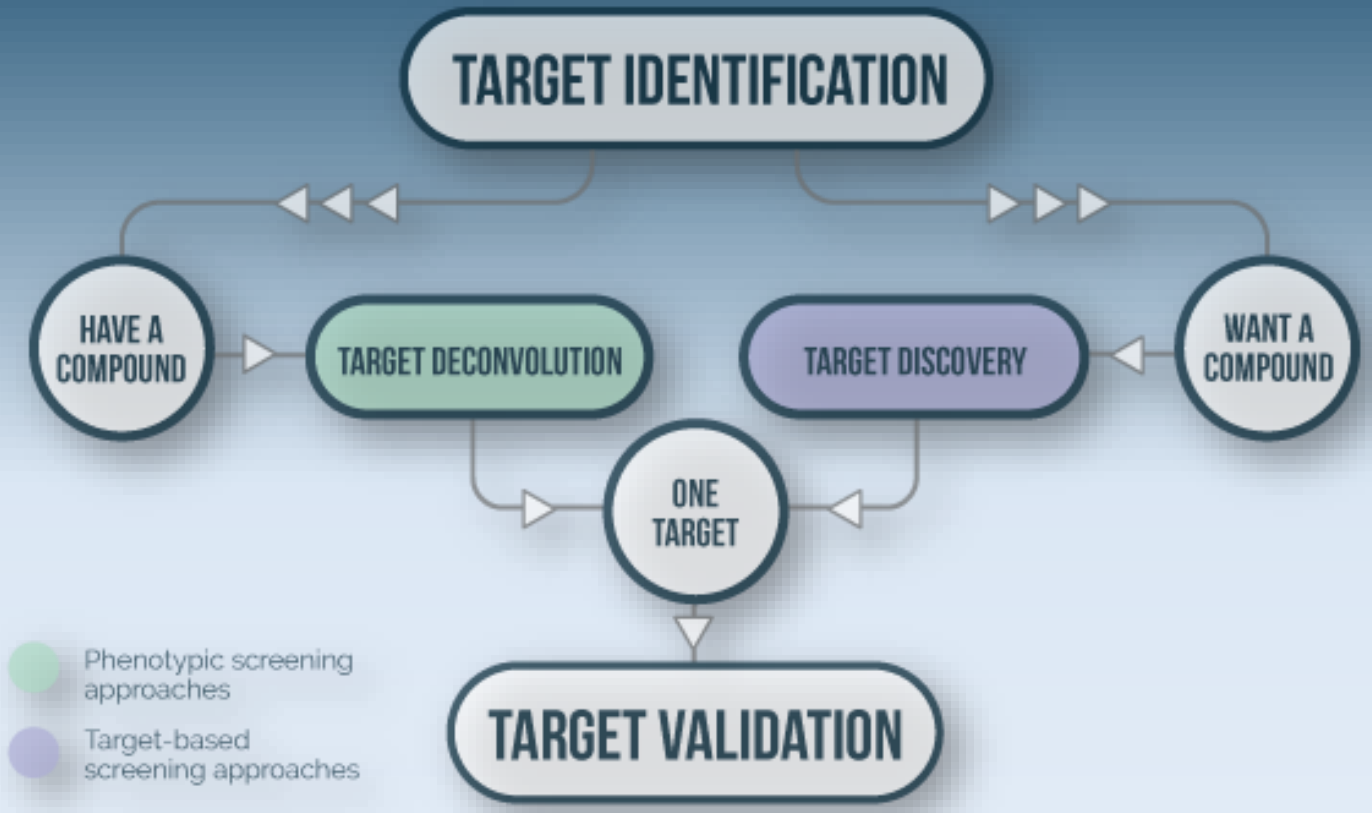
고려대학교
KOREA UNIVERSTY

# Answer to the Last Class

- 중간고사 결산
- 중간고사 정답 설명



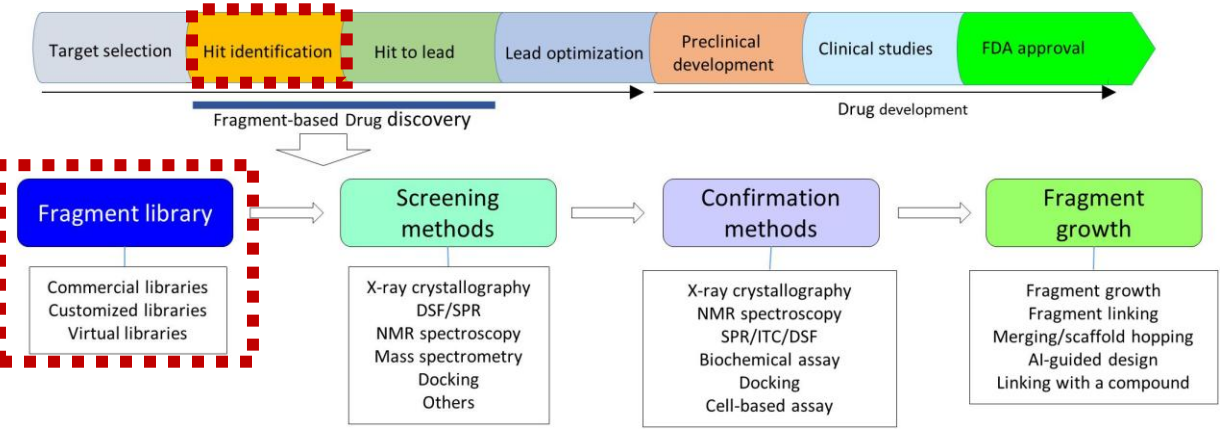
# Questions?

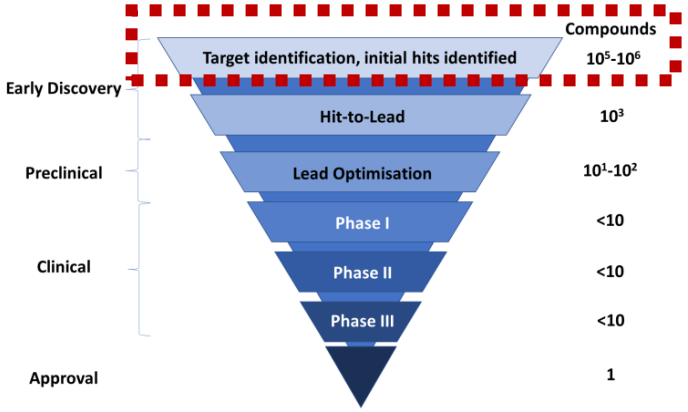



# Chapter 3-2.
## 후보물질 탐색 및 발굴

# 4. Hit (유효물질) 도출 (Screening/Target to Hit)

- Target 이 검증되고 TPP 작성이 완료되면, 간단하면서도 선택성이 높은 "Screening"법을 선택하여 유효물질 (Hit)을 도출하여야 함.
  - Target에 반응하는 유효물질 (Hit)을 선별하는 과정
  - 합성신약의 경우, 타겟 단백질 어세이를 개발해서 자동화된 High throughput screening (HTS) 통해서 수십만에서 수 백 만개의 화합물을 스크리닝 가능
    - Commercial library를 구매하거나, 실험실/제약회사가 소유한 customized library를 이용할 수 있음
    - 최근엔 Computing 기술에 기반한 Virtual library 를 통한 screening 법도 개발됨
  - 합성된 화합물의 효능을 단시간 내에 검색할 수 있는 동시대량효능검색법 기술의 발달로 합성된 화합물의 효능을 동시 검색할 수 있음
  - Hit 화합물을 선정하는 특별한 기준은 없으나 합성신약의 경우 10 uM- 수십 uM 이하의 농도에서 Target과 반응하는 화합물을 선별
  - 또한 후속연구를 진행하기 위하여 화학구조가 정확한지 확인

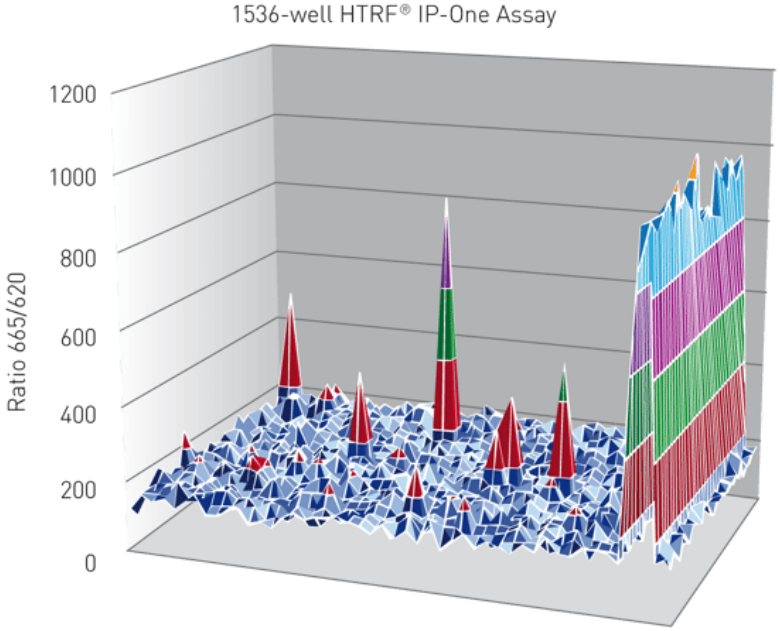


# High Throughput Screening (HTS)



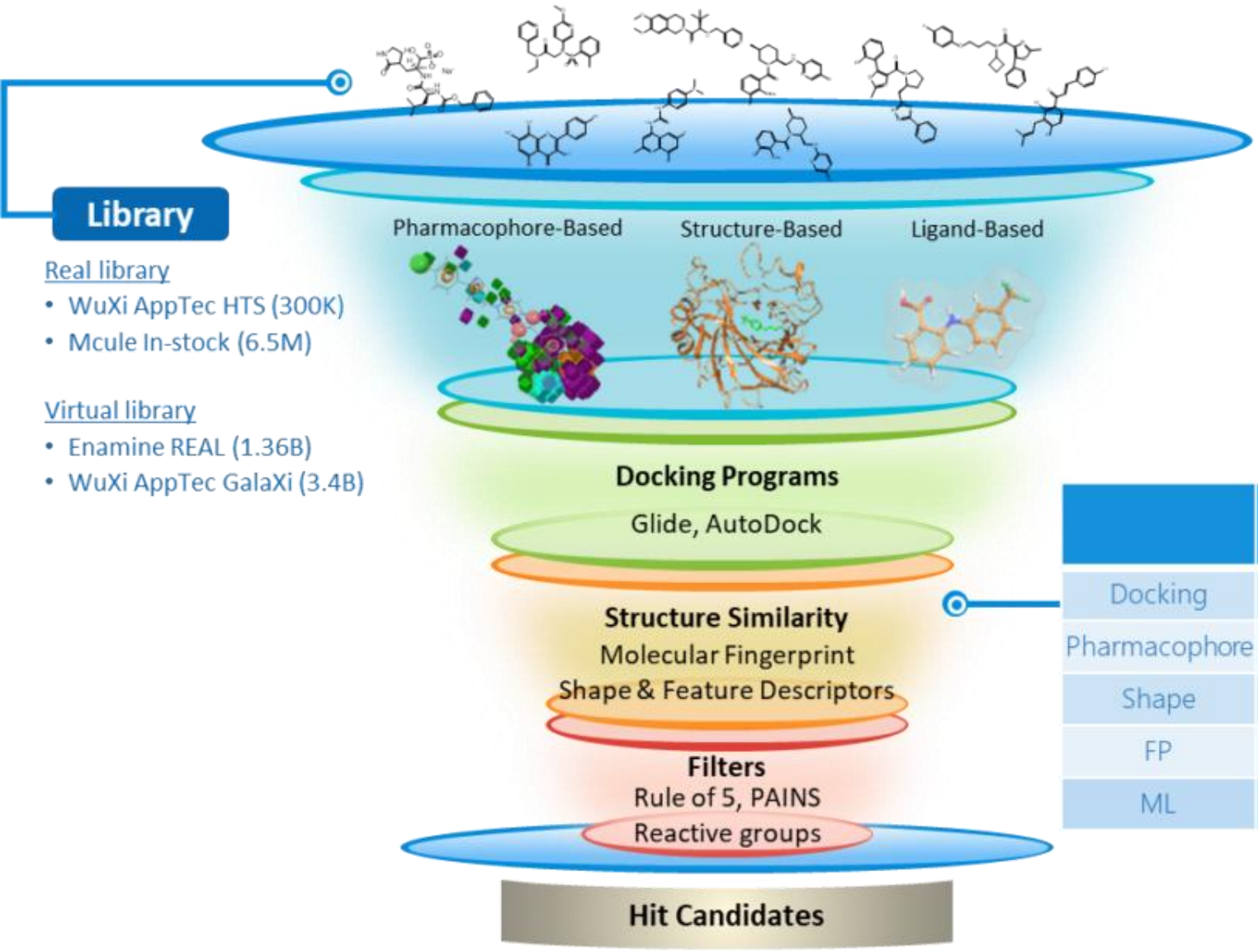


# High Throughput Screening (HTS)

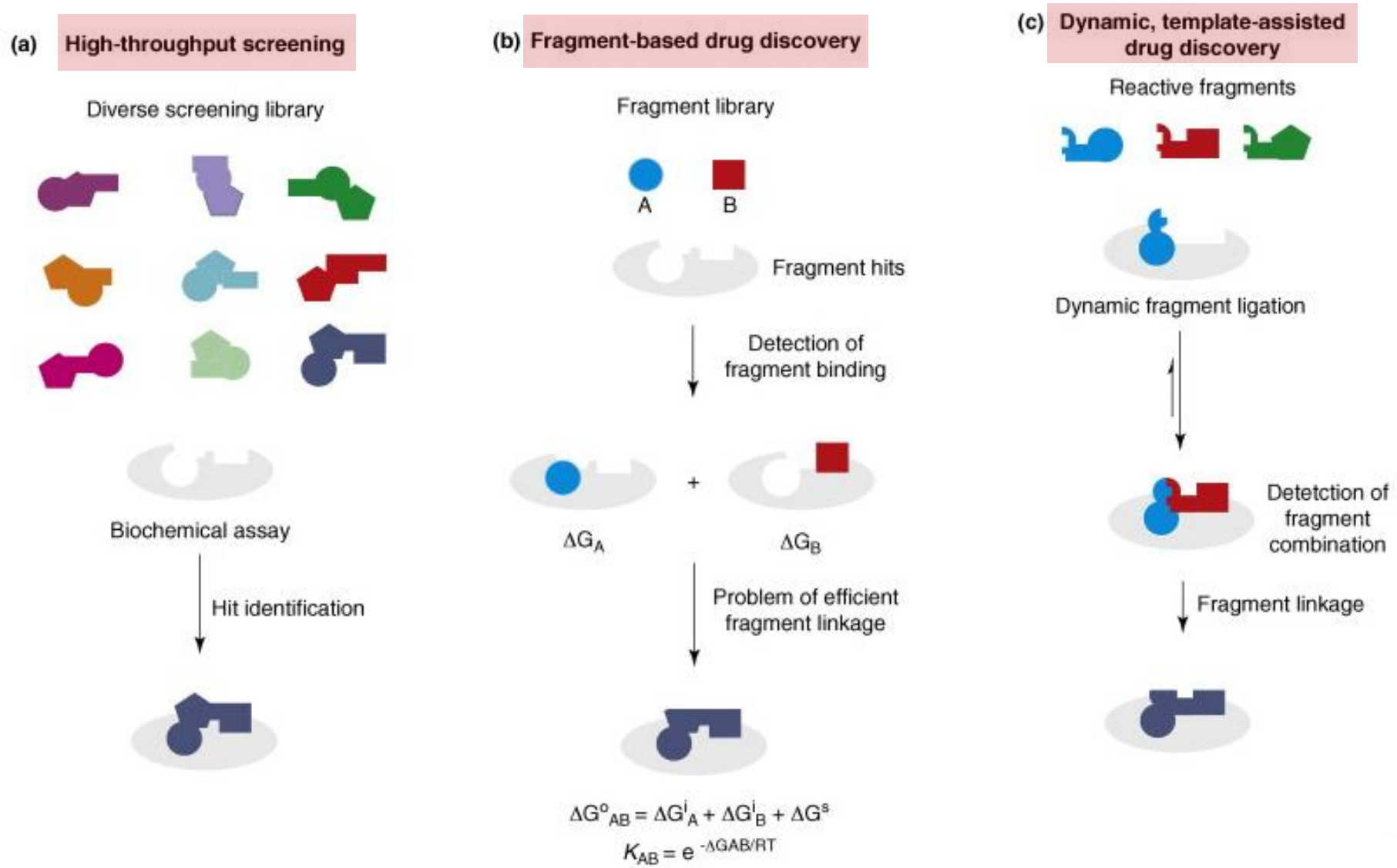


# 4. Hit 도출을 위한 Screening

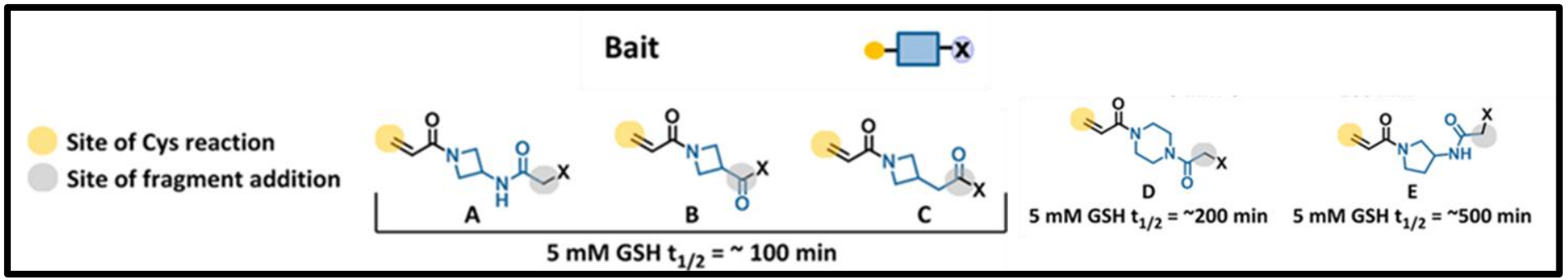


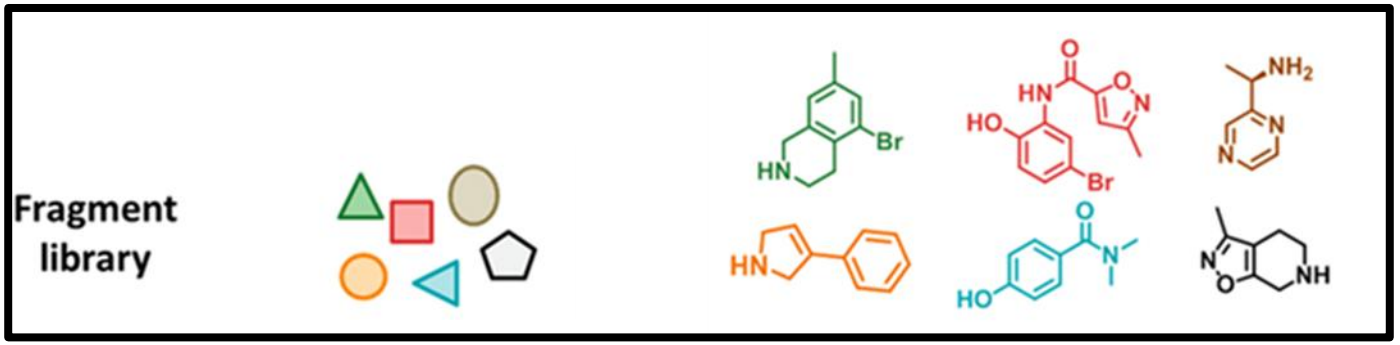

| | Chemical Space | Time consuming |
|---|---|---|
| Docking | Millions | Days |
| Pharmacophore | Millions | Days |
| Shape | Billions | Hours |
| FP | Billions | Hours |
| ML | Billions | Hours |

# 4. Hit 도출을 위한 Screening



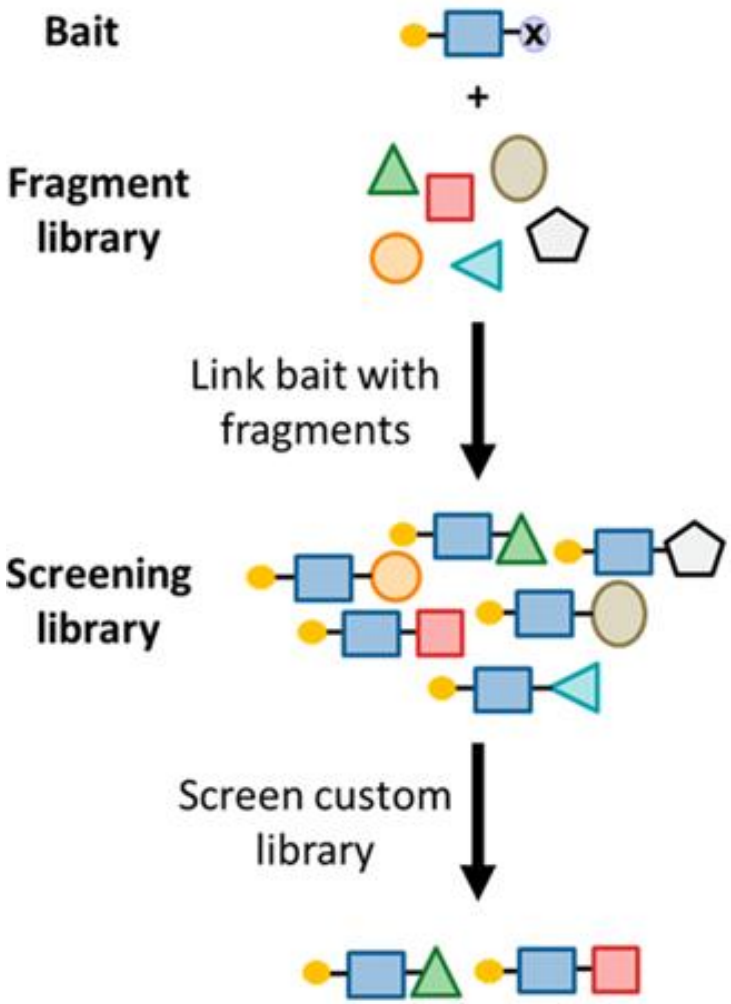
# 4. Hit 도출을 위한 Screening

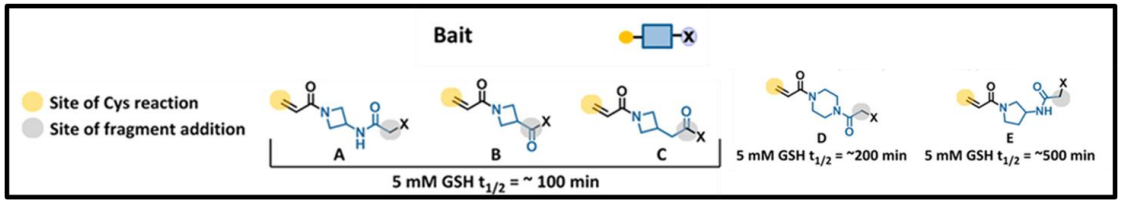


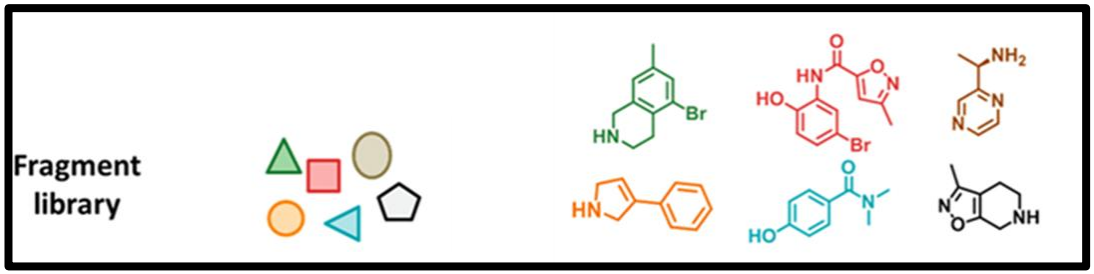




**Hit (유효물질)**

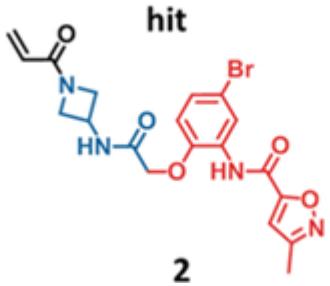
# 4. Hit 도출을 위한 Screening



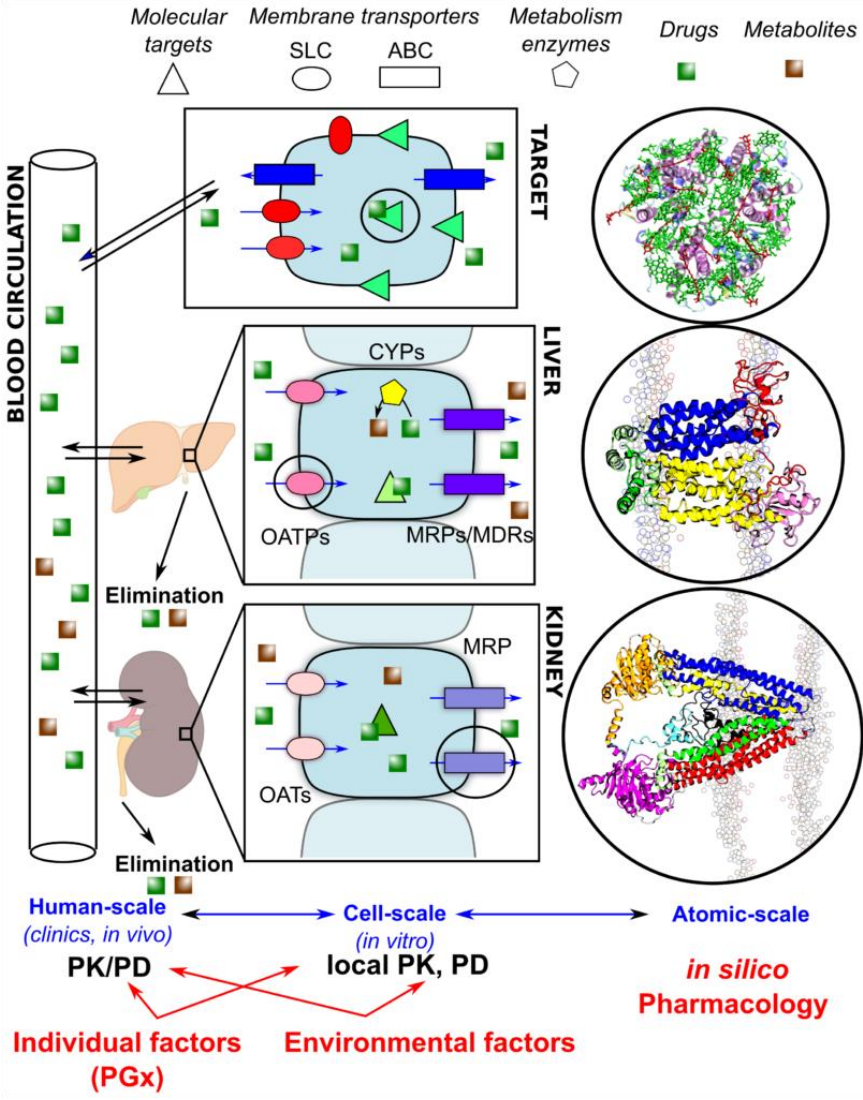




**3300 molecules made and screened**
**8 selective hits confirmed by MS**

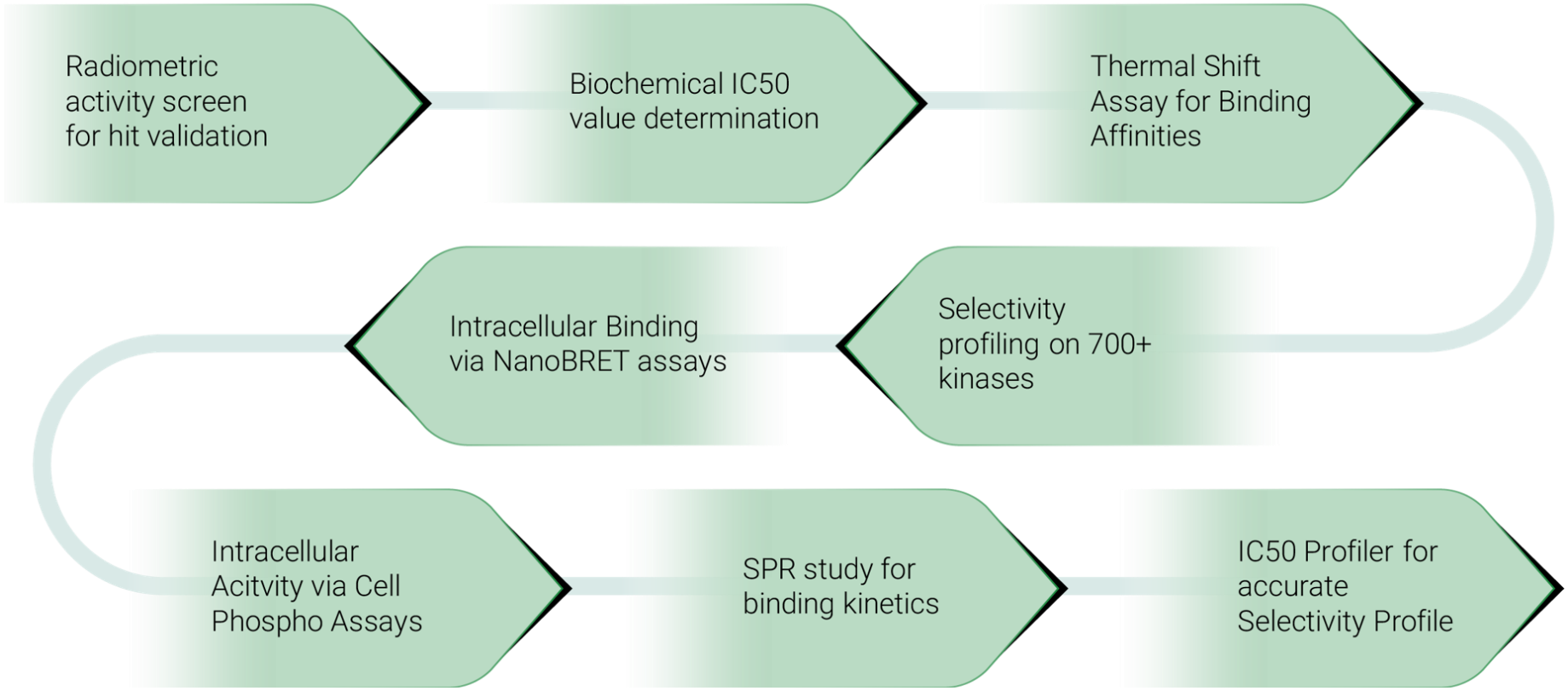


# 4. Hit (유효물질) 도출 (Screening/Target to Hit)

- 연구 초기에는 모두 확인하기 어려우나 Pharmacokinetics (PK) data [ADME]를 *in silico* [컴퓨터 모의실험 또는 가상실험]에서 수행하여 개발 가능성이 있는지에 대한 검증도 수행
- 생물학적 타겟에 따라 기존에 확보하고 있는 화합물 library 수천 개를 대상으로 다양한 검색을 하여 보통 여러 개의 1 차 hits (primary hits)를 선별
- Lead 화합물의 효과적인 발굴을 위하여 화학 구조적 다양성을 갖는 3~5 종 이상의 유효물질 (Hit) 확보 필요.
- 1 차 hits 는 활성과 관련, 제한된 data 만 확보한 상태로 생물학적인 활성을 검증하는 단계를 거쳐야함
- Hit 화합물이 발굴되면 combinatorial chemistry 등과 같은 고속합성기술을 이용하여 hit 의 다양한 유도체를 합성하여 더욱 효과가 좋은 lead 를 도출

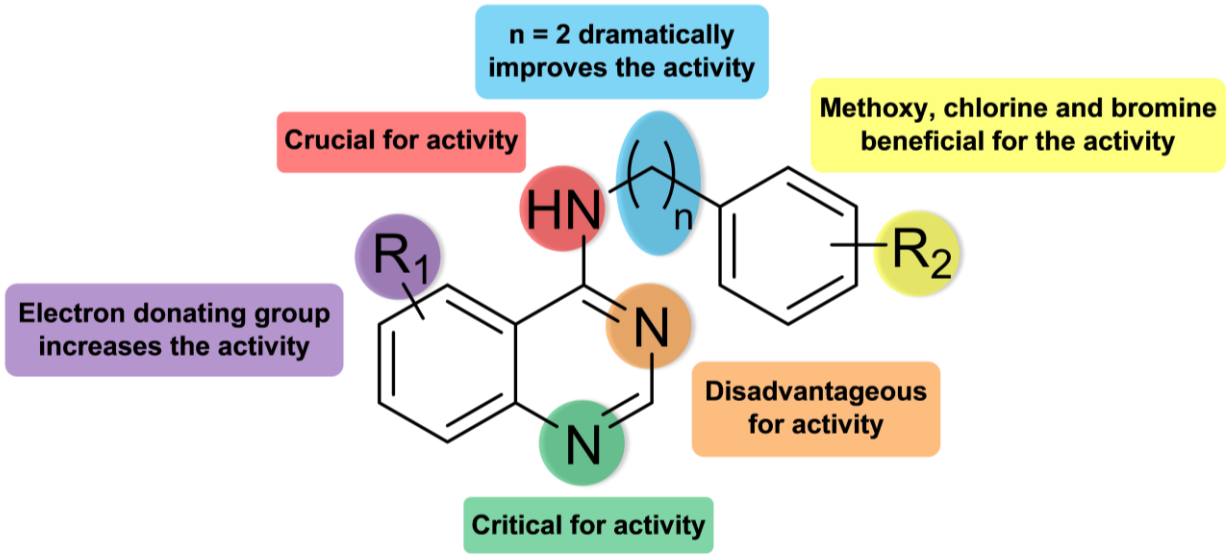


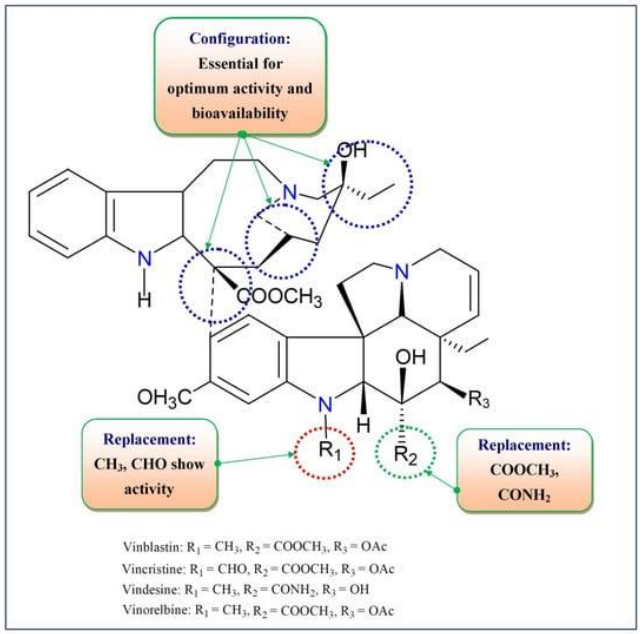
# Screening options for hit identification



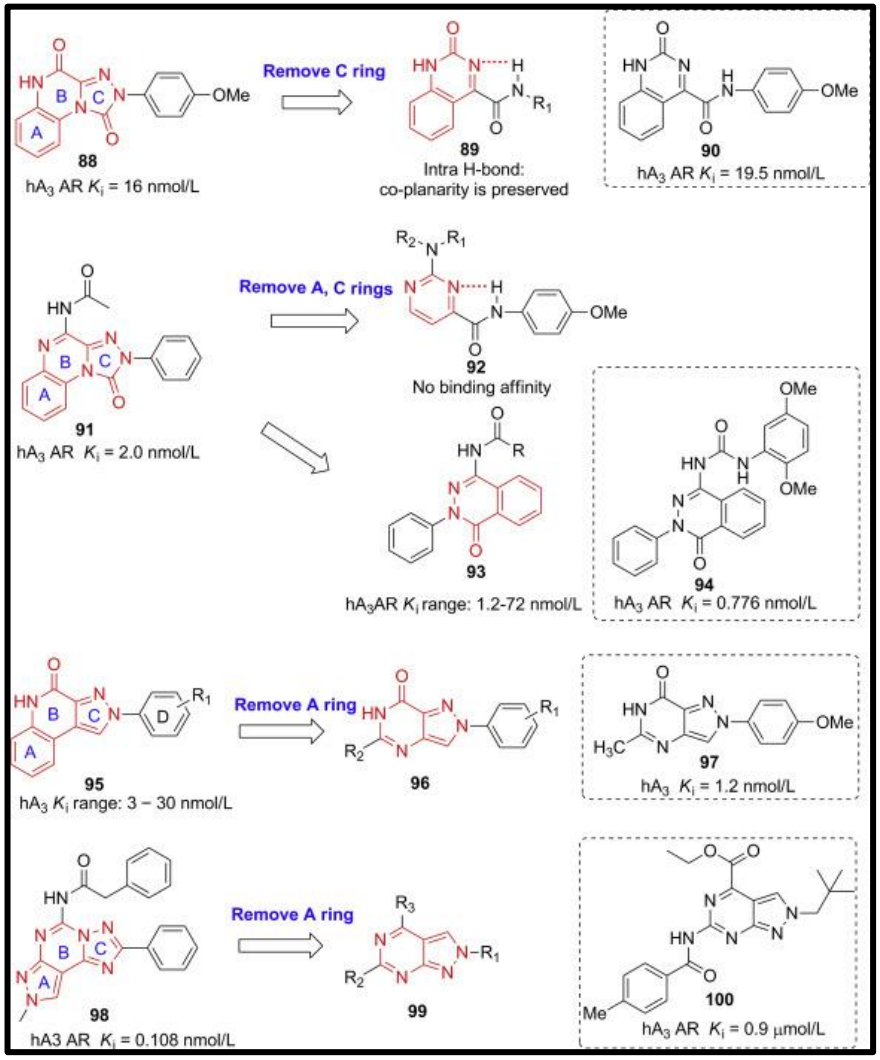
# 5. Lead (선도물질) 도출 (Hit to Lead)

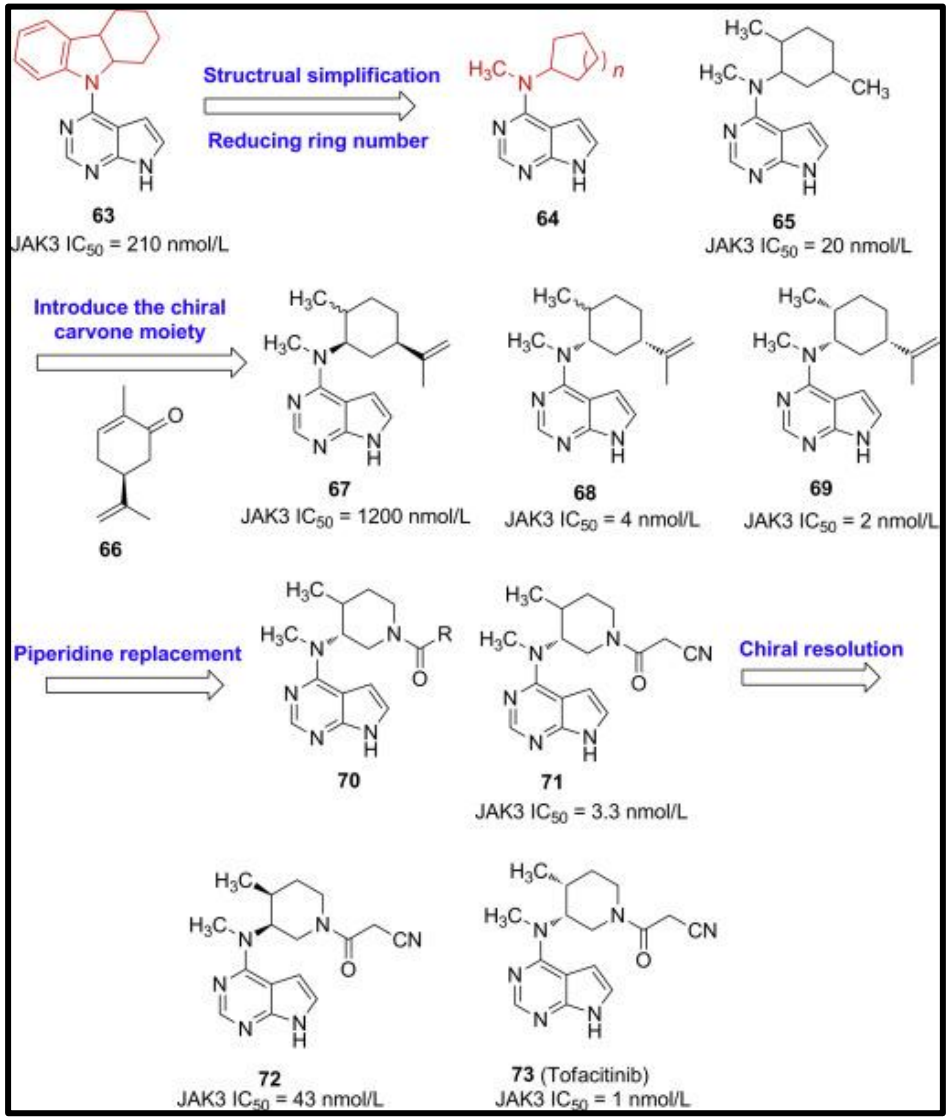
- Lead 도출 및 최적화: Hit (유효 물질) 중 가장 효과적인 물질을 찾는 선도물질 (Leading compound) 연구 단계
- 연구자의 경험을 이용한 SAR (Structure Activity Relationship) 연구를 수행하여 hit 화합물 개량
  - 예를 들면 hit 화합물의 구조를 몇 개의 부분으로 나눔
  - 각 부분에 다양한 치환기를 도입하는 가설을 세운 후 활성평가를 수행
  - 가설의 검증을 수행하는 방식으로 진행
- Hit 에서부터 Lead 를 도출하는 단계에서 진행된 SAR 연구를 통하여 화학구조와 target 에 대한 상관관계를 규명함으로써 구조를 최적화해 나간다.



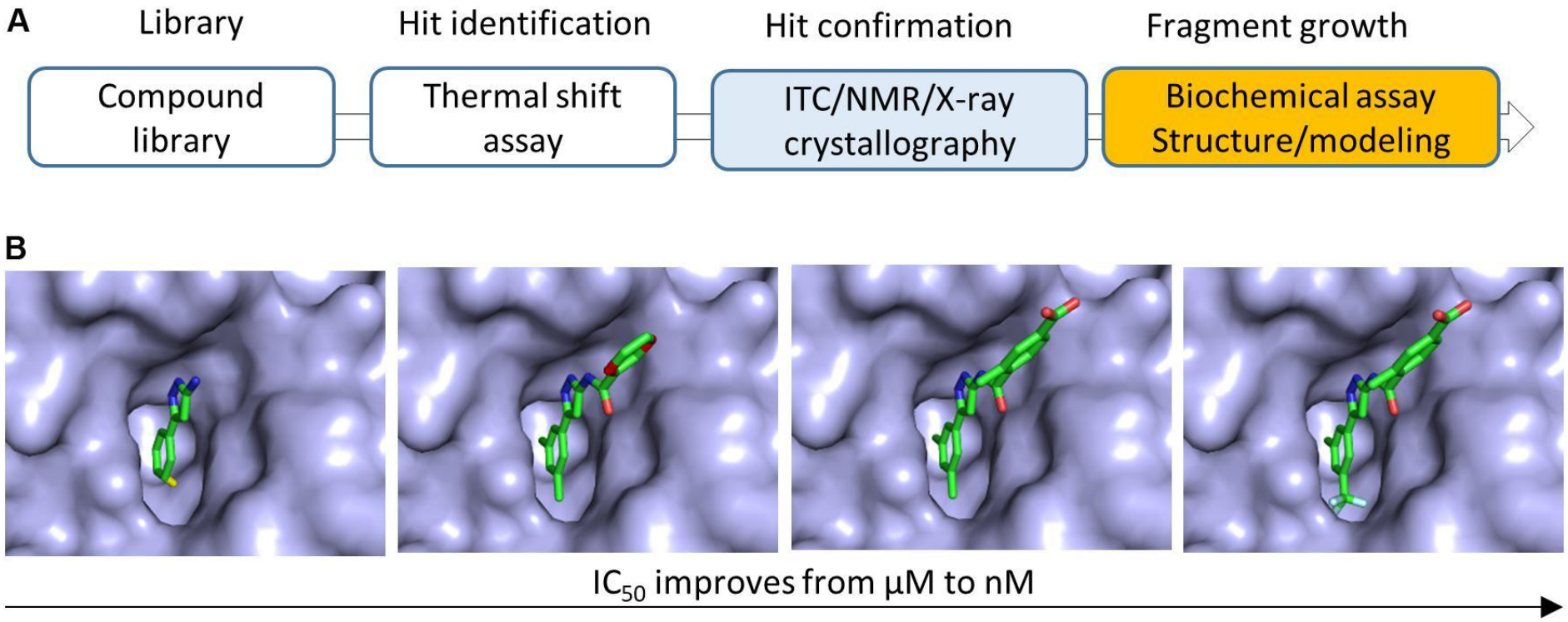



# 5. Lead (선도물질) 도출 (Hit to Lead)



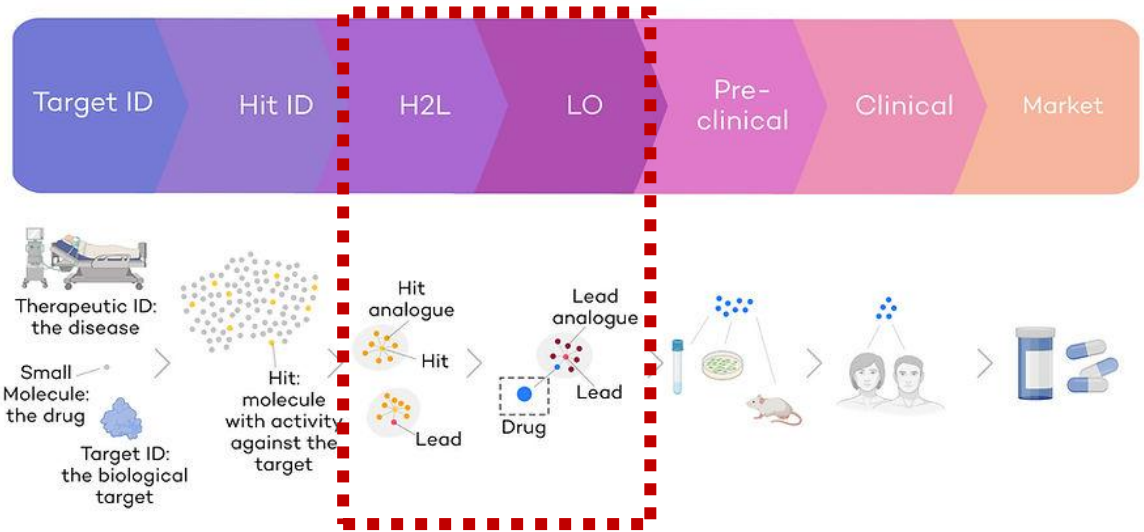



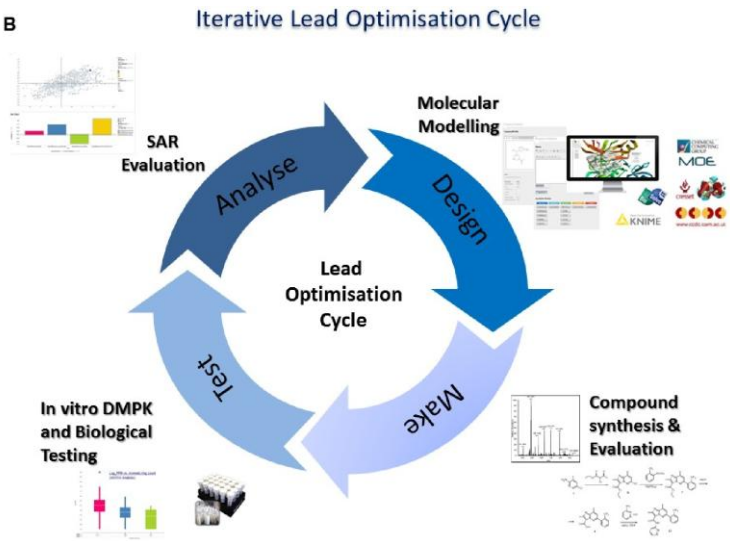
# 5. Lead (선도물질) 도출 (Hit to Lead)



# 5. Lead (선도물질) 도출 (Hit to Lead)

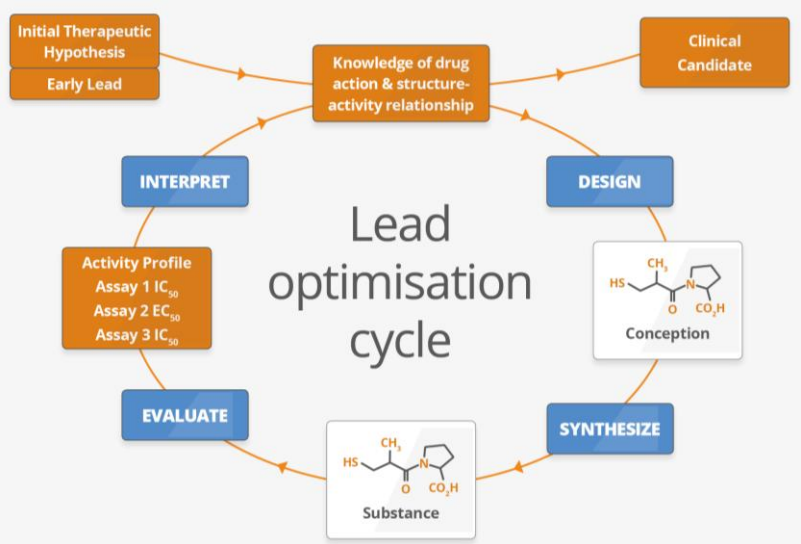
- 이 과정에서 hits 는 보다 강하고 (potent) 선택적 (selective)으로 작용하고 적합한 in vivo models 에서 약효 (efficacy)를 나타낼 수 있는 PK 특성을 보유하고 있는지 평가
- 이때 각 핵심구조 부위에 대해 SAR 연구를 집중하고 각 compound 의 효능과 선택성을 확인.
- Lead 화합물은 정확한 정의는 없으나 다음 정도의 화합물을 말한다.
  - 가. 수 백 nM 수준의 효과를 나타낸다 (Hit의 경우 uM 수준)
  - 나. 타겟에 대한 유효성이 어느 정도 확인되었다
  - 다. Mechanism of action (reversible/irreversible, kinetics) 이 확립되었다
  - 라. 다른 분자타겟에 작용하는지 선택성을 확인하였다.
  - 마. 경쟁력 또는 신규성을 포함한 특허성을 확인하였다.
  - 바. Preliminary PK(약동학: 약의 용량과 체내의 약물농도간의 관계) in vivo (rodent)

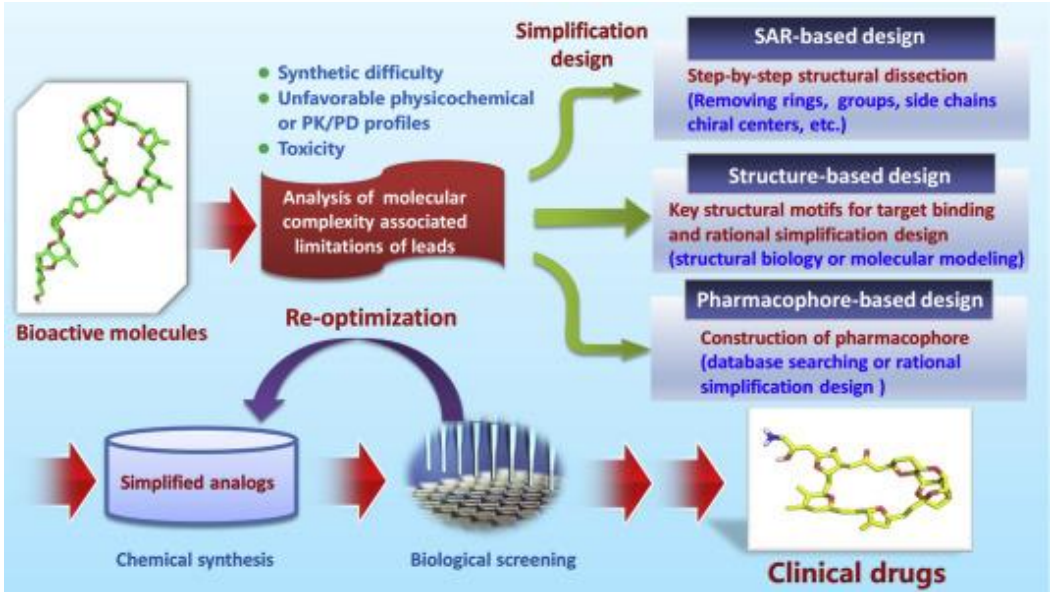


# 6. Lead Optimization

- Lead optimization is a critical process that culminates in the identification of a preclinical candidate. The most promising hit series, once they are identified through hit-to-lead efforts, advance into the lead optimization stage of drug discovery.
- Lead compound 가 선정되면 Lead compound 의 유도체 합성을 통하여 최적의 활성을 갖는 물질들을 확보하고 SAR(structure-activity-relationship) 확립
- 이 단계에서 기본적인 약리 및 초기 스크리닝 독성시험과 ADME study 를 통하여 후보물질을 선정
- Lead 최적화의 목적은 lead 화합물들의 우수한 특성은 유지시키면서 lead 구조에서 단점은 개선시키는 것

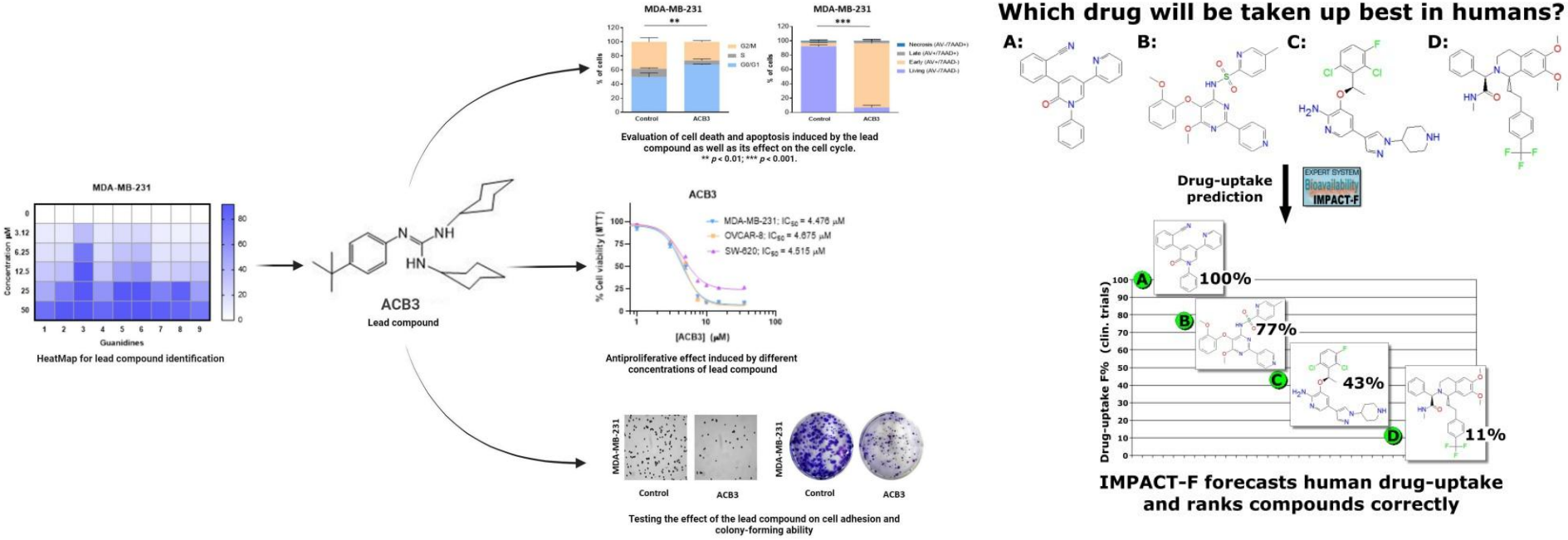


# 6. Lead Optimization

- Lead 의 potency 및 efficacy 를 개선하고 타겟 외의 활성을 감소시키거나 배제시키며 아울러 선택성, 대사 안정성 (metabolic stability) 등을 최적화
  - 예: 초기 lead 화합물들 보다는 부작용을 나타내는 화합물의 target affinity는 최소화시키고 흡수는 개선시키기 위하여 target affinity 와 장세포 투과도 시험 진행

  → 타겟에 대한 이들 물질들의 효력 (potency) 와 선택성은 유지 + 부작용 affinity는 감소, 장투과도는 개선된 물질들을 찾아냄.



# 6. Lead Optimization

- Lead 화합물 최적화를 통한 Candidates 선정은 hit-to-lead 과정과 유사하며, candidate 는 정해진 조건은 없으나 다음의 특징을 갖는다.

  가. 100 nM 이하의 EC50 (최대 효과의 50%를 얻는 데 필요한 용량)

  또는

  IC50 (특정 억제 물질이 시험관 내에서 주어진 생물학적 성분을 50% 억제하는데 필요한 양)

  나. 유사한 타겟과 1,000 배 정도의 선택성

  다. 적절한 용해도

  라. 적절한 in vivo PK parameter

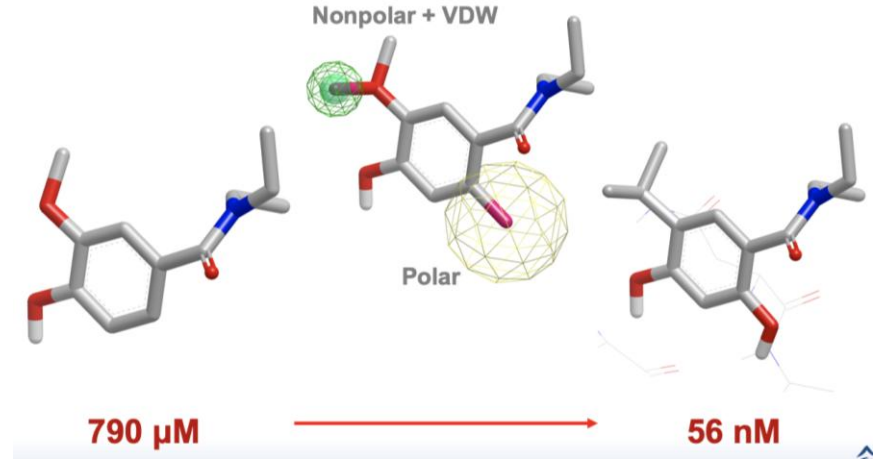
  마. 주요 CYP (cytochrome P450: 약물대사 효소) 에 대한 inhibition 또는 induction 확인

  바. 적합한 동물 모델에서의 약효

  사. 단회투여 또는 2 주 정도의 독성시험에 대하여 안전성 확보

  아. In vitro mutagenicity(Ames fluctuation test, in vitro Micronucleus test)

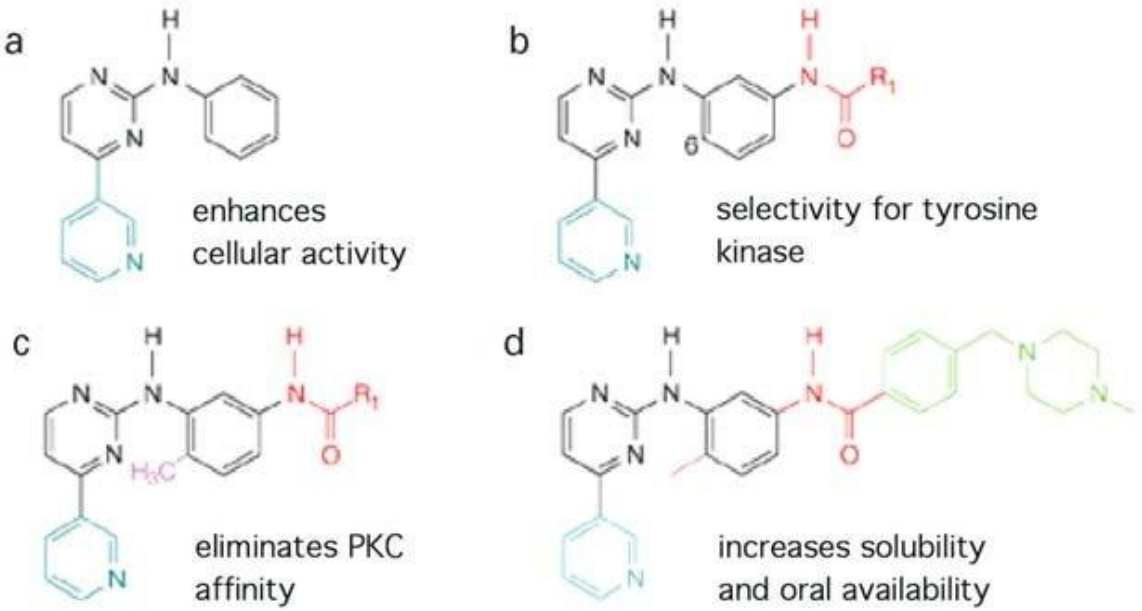
  자. 특허 출원

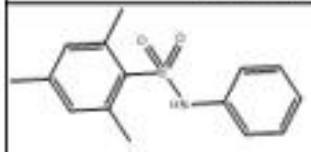
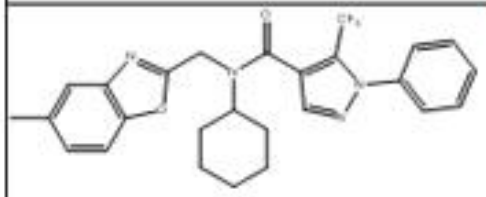





# 6. Lead Optimization

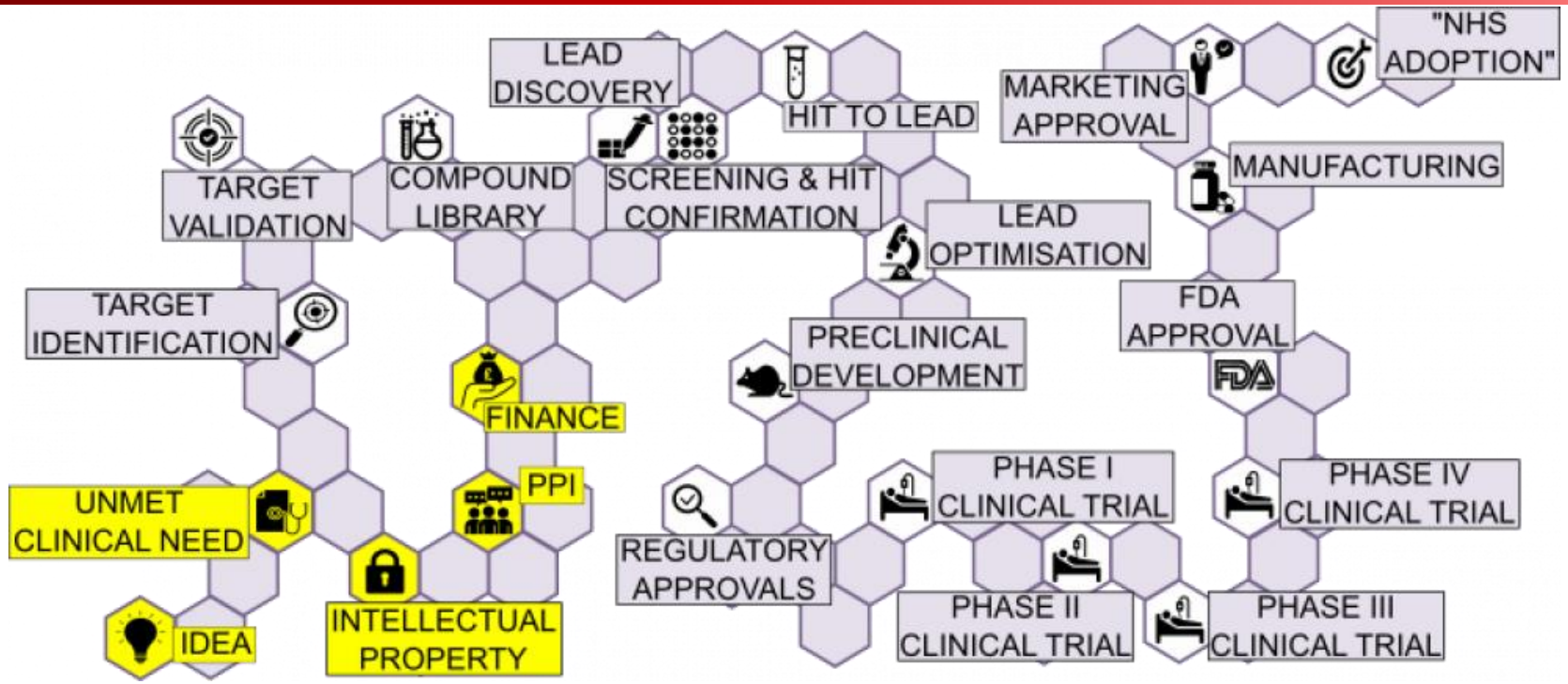
## Lead optimization: Glivec® example




| STRUCTURE | OBJDID | MW | TPSA | Rotatable Bonds | cLogP | Kinetic Solubility (µM) | Primary Assay Activity (nM) | 1° assay/selectivity assay (IC50 data ratio) | Ligand Efficiency | Score |
|---|---|---|---|---|---|---|---|---|---|---|
|  | CMPD_1 | 275.37 | 42.22 | 3 | 2.74 | 116 | 413 | 177 | 0.52 | 0 |
|  | CMPD_2 | 482.50 | 62.79 | 4 | 5.72 | Not Acquired | 212 | 363 | 0.29 | 5 |

# UCL (University College London)

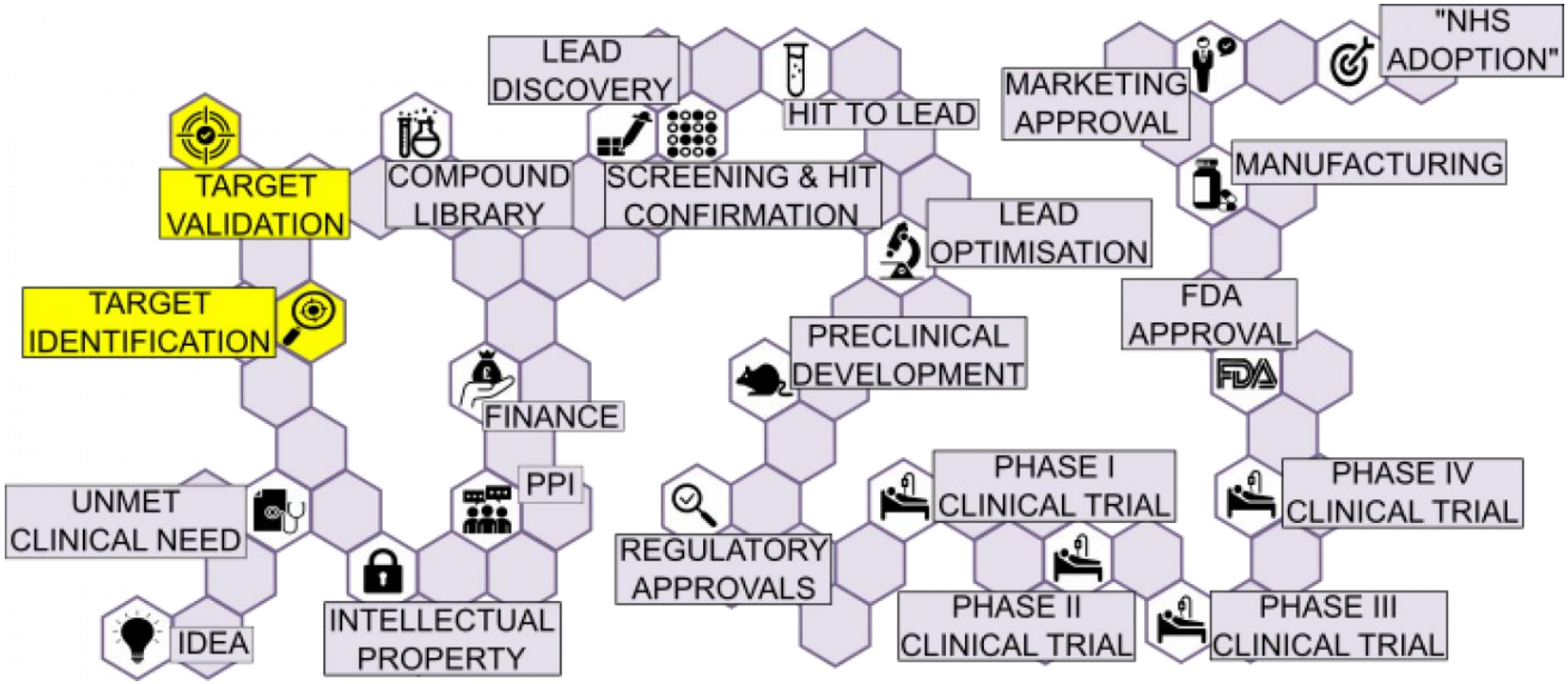
## Small Molecules Roadmaps




- IDEA: Target Product Profile (TPP) with UNMET CLINICAL NEED
- INTELLECTUAL PROPERTY
- PATIENT & PUBLIC INVOLVEMENT (PPI): Understanding the patient and clinician needs - Does your TPP meet all of the essential requirements expressed by frontline clinical care providers, patient and public contributors (PPI)?

# UCL (University College London)

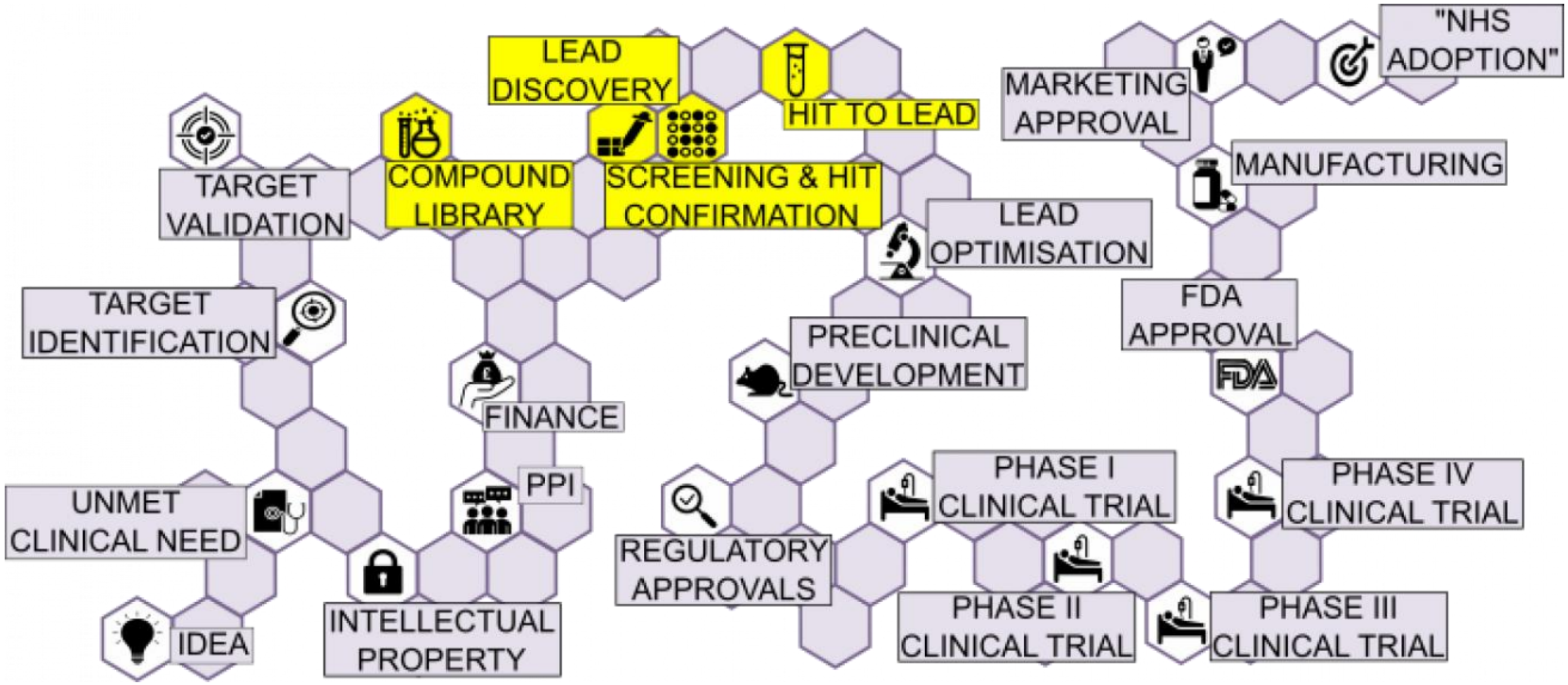
## Small Molecules Roadmaps




- TARGET IDENTIFICATION: Identifying biological molecular structures which are 'Druggable'. Druggable biological targets interact with the small molecule to change or alter the function of the target.
- TARGET VALIDATION: A prospective drug target has to undergo many validation experiments to show that it is directly involved in the pathway and that it will have a therapeutic effect.
- ASSAY VALIDATION: After identifying a target, a simple experiment to analyse enzyme activity or cell phenotype must be generated to screen against a small molecule library. This 'Primary assay' will be used to screen as many compounds as possible – potentially millions – so must meet a assay validation threshold before being used.

# UCL (University College London)

## Small Molecules Roadmaps

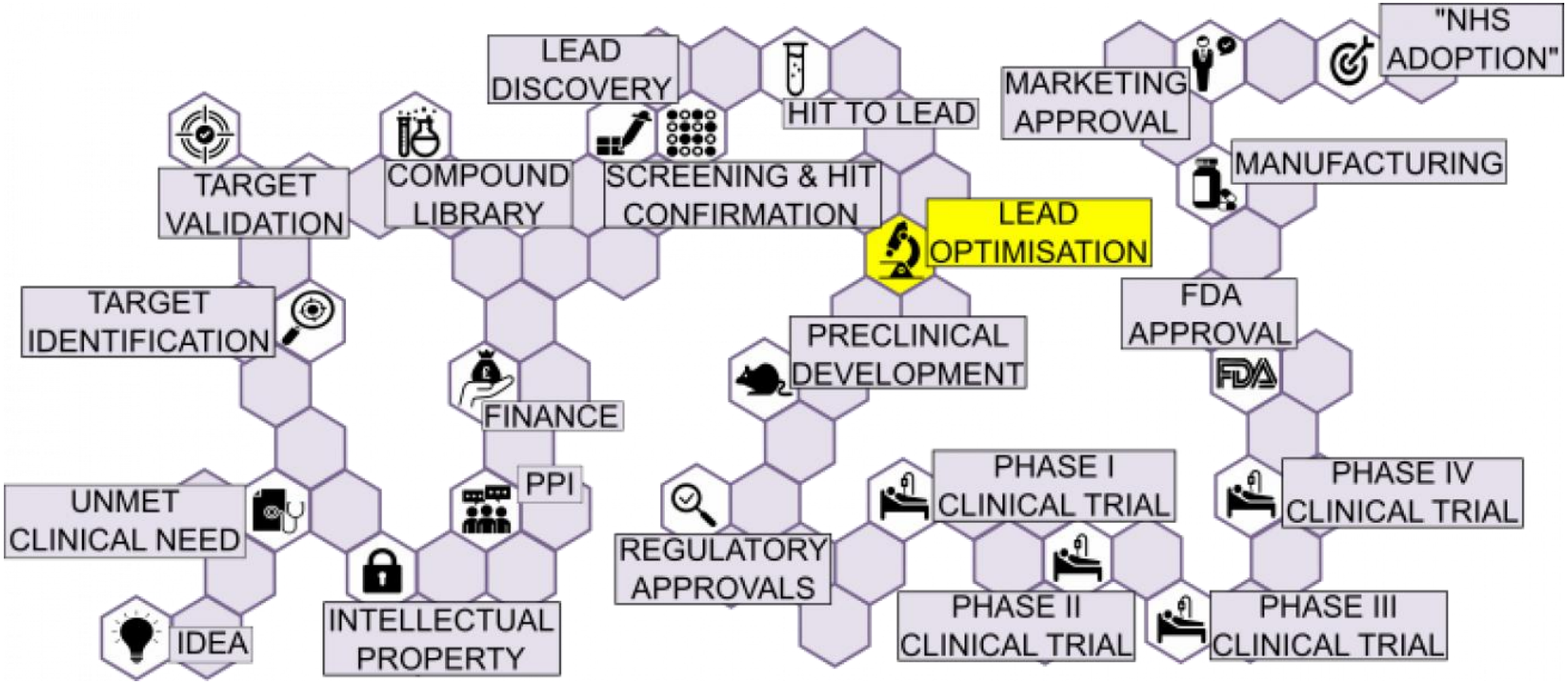



- SCREENING & HIT CONFIRMATION: Once a target is known and an assay for activity has been developed, the medicinal chemist search for compounds that can potentially interact with the target. The first step is to find a 'hit', which means poorly optimised compounds that nevertheless specifically interacts with the target.
- HIT TO LEAD: Hits compounds are refined into a short-list of Lead compounds, depending on the following factors:

•Synthetic accessibility
•Potency — Dose response
•Specificity
•Selectivity
•Non-toxic
•Drug-like structure
•Preliminary Structure Activity Relationship (SAR)
•Physical properties — Solubility — Lipophilicity

# UCL (University College London)

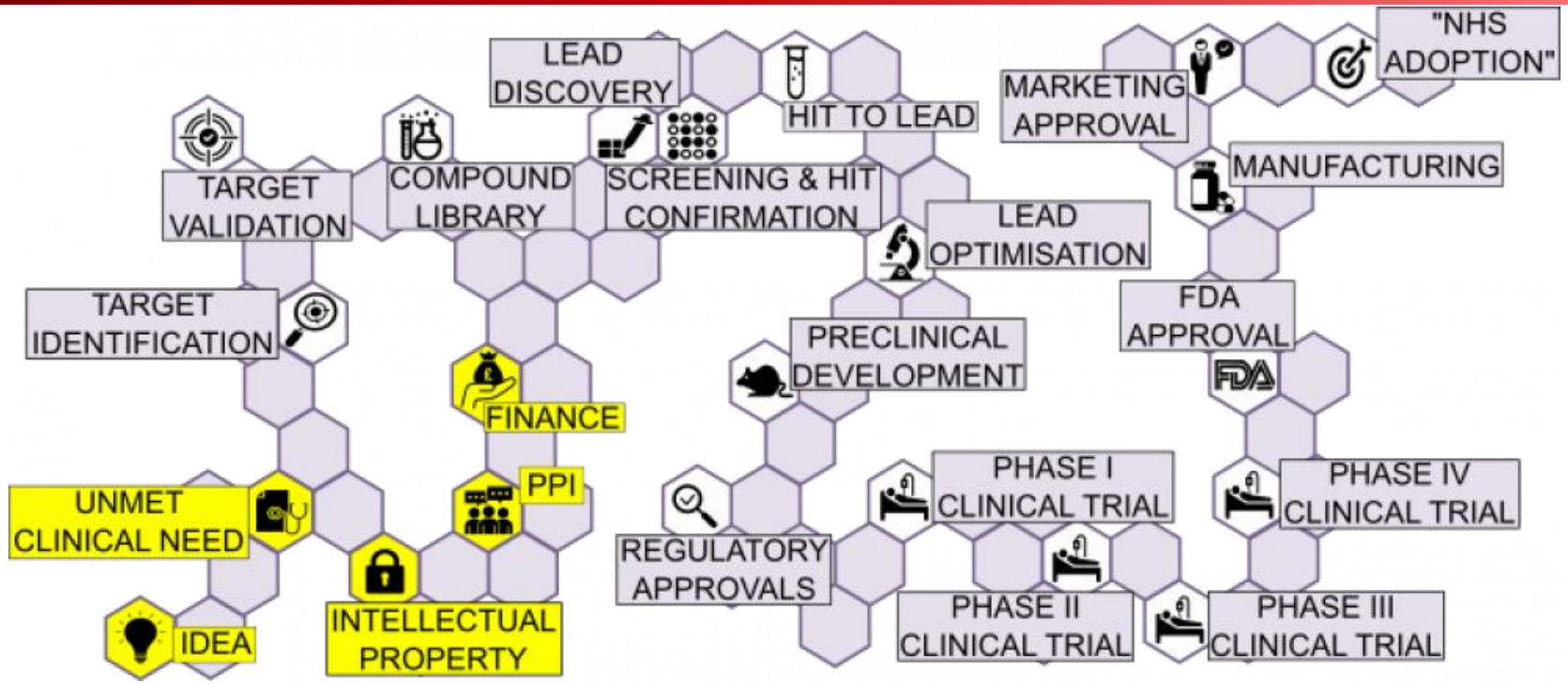
## Small Molecules Roadmaps




LEAD OPTIMISATION: The aim in lead optimisation, is to improve biological activity, target selectivity, potential for toxicity. Finally the compounds will be evaluated for how they are absorbed, distributed, metabolised and eliminated when dosed in an animal model of disease. Additionally the efficacy of the compound to modulate the disease will be evaluated in the animal model.

# UCL (University College London)

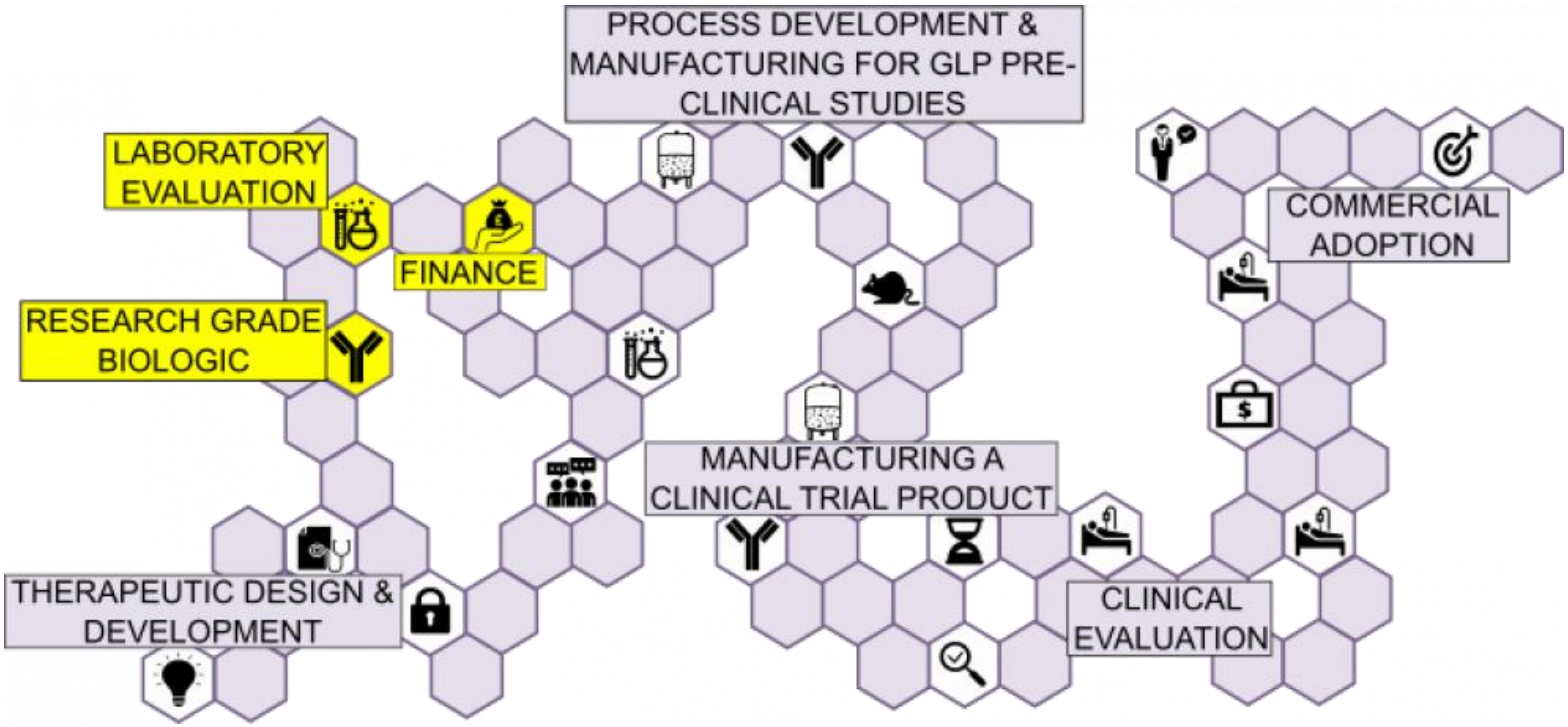
## Biologics Roadmaps




- IDEA: Target Product Profile (TPP) with UNMET CLINICAL NEED
- INTELLECTUAL PROPERTY
- PATIENT & PUBLIC INVOLVEMENT (PPI): Understanding the patient and clinician needs - Does your TPP meet all of the essential requirements expressed by frontline clinical care providers, patient and public contributors (PPI)?

# UCL (University College London)

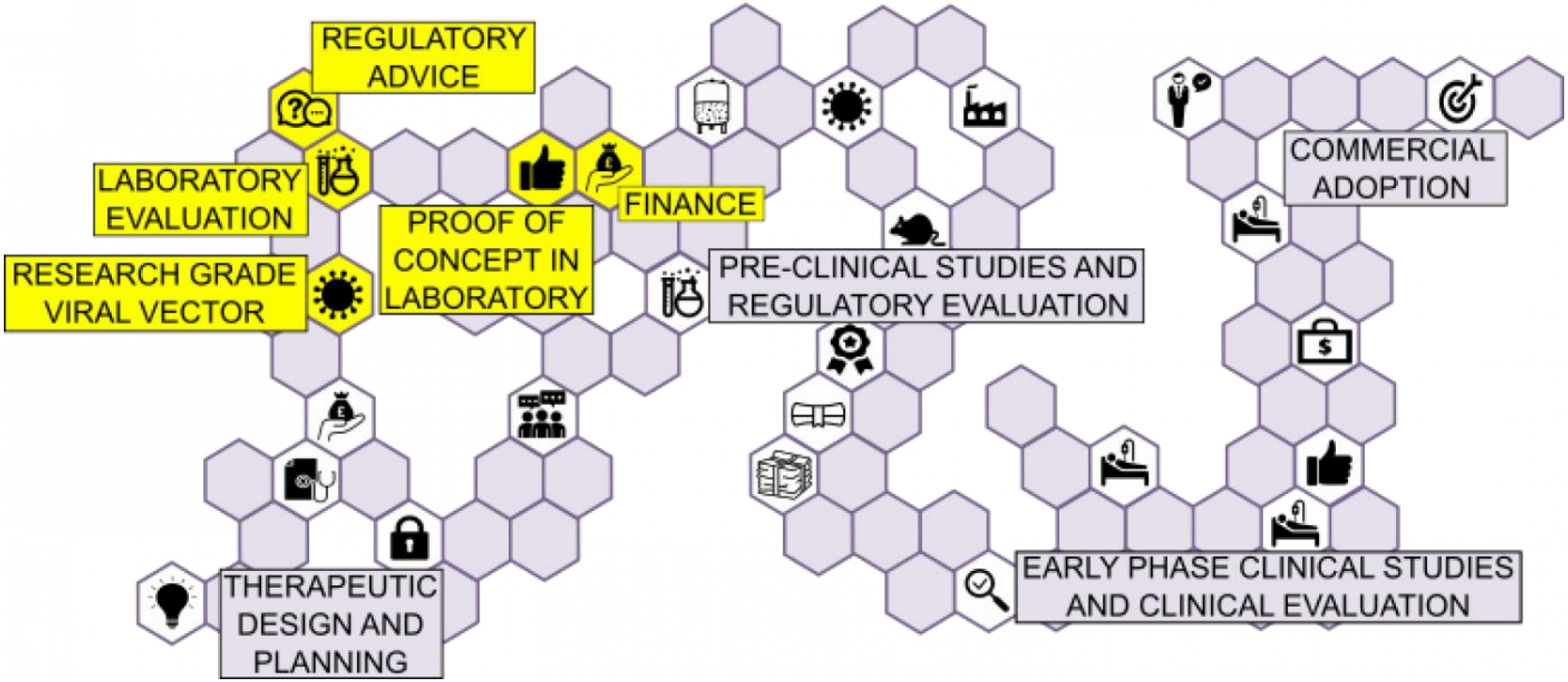
## Biologics Roadmaps




- RESEARCH GRADE BIOLOGIC: Align form of biologic(proteins, antibodies (appropriate IgG backbone) or derivatives, peptide, oligonucleotides, vaccines) to indication/route of administration/immune response needs.
- Considerations : Production/selection "clones/cell lines", Mammalian/bacterial production
- LABORATORY EVALUATION: Laboratory evaluation is a crucially important step to develop early Proof-of-Concept and secure follow-on funding.
- FINANCE: Regardless of altruistic drivers, development of biologics has significant costs that must be met and VAT needs to be added for manufacturing and/or services not resulting in a medicinal product.

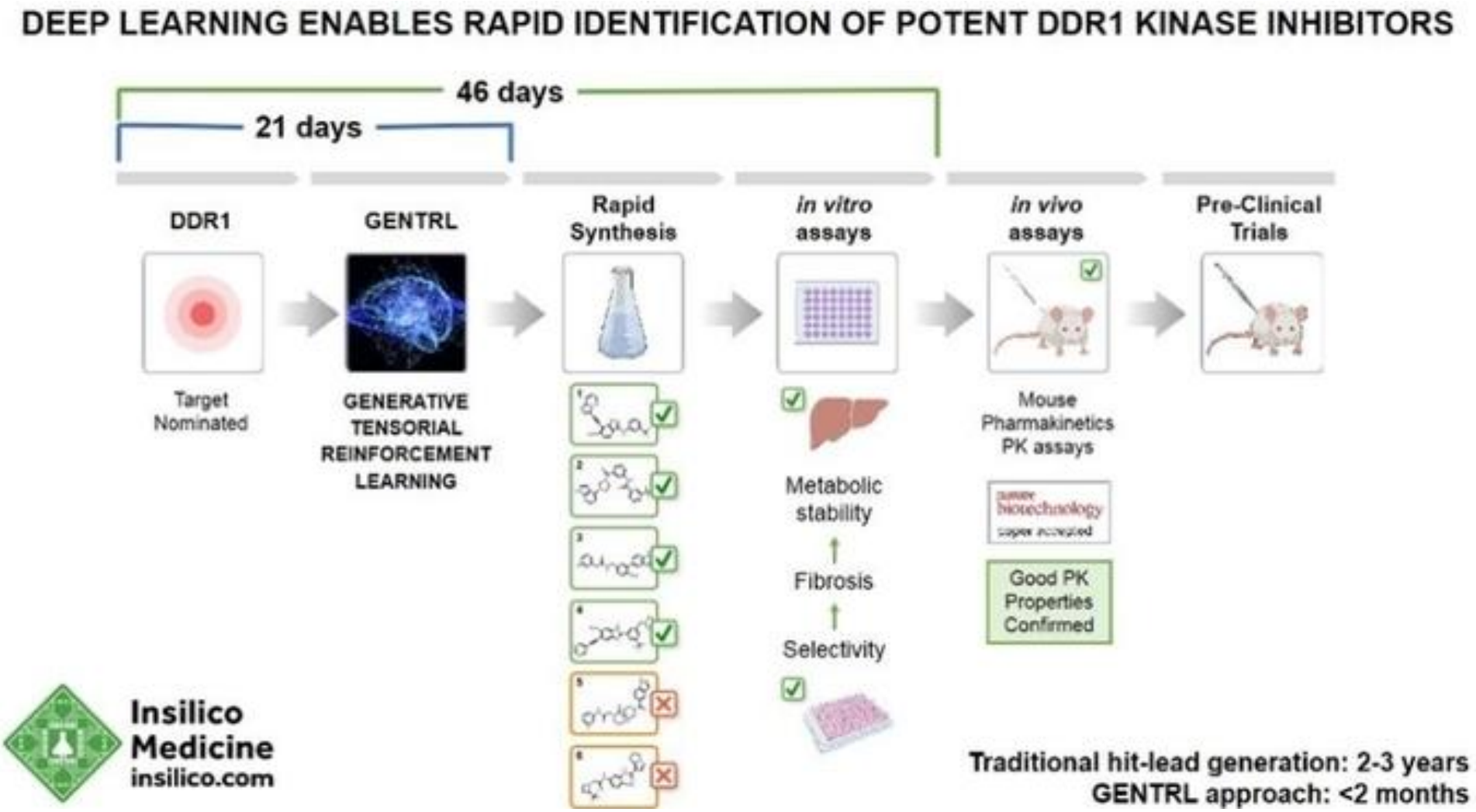
# UCL (University College London)

## Cell & Gene Therapy Roadmaps

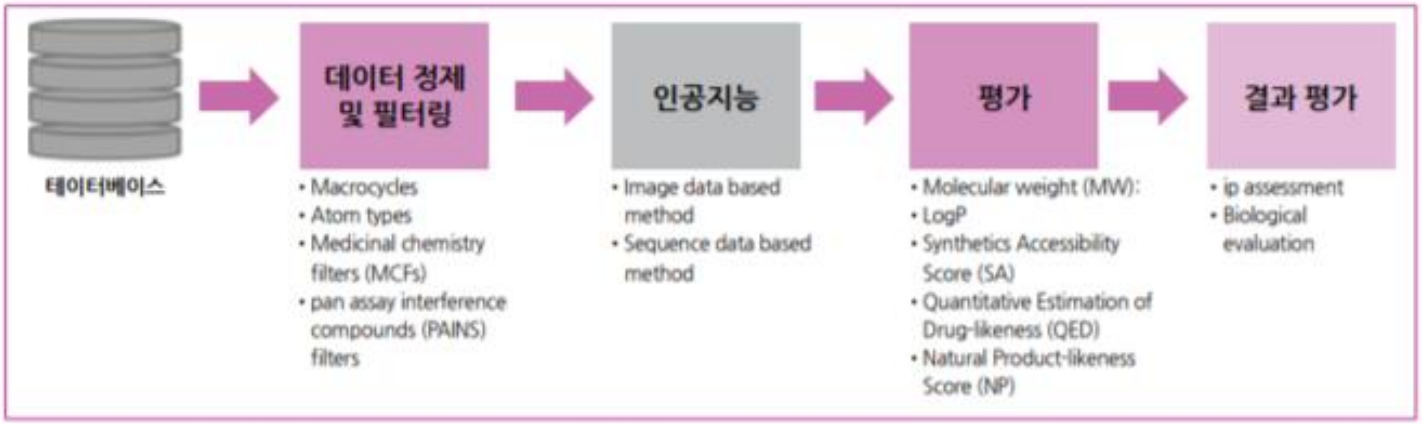



- RESEARCH GRADE VIRAL VECTOR: In order to develop a research grade cell and gene therapy vector, there are a number of factors to consider in order to ensure the vector is stable, sufficient and optimised for preclinical evaluation.
- LABORATORY EVALUATION: Once the research grade product has been developed, it is important to ensure that it matches the Target Product Profile (TPP) and meets all essential requirements.
- REGULATORY ADVICE: Seek regulatory advice to ensure that your project is compliant with regulatory guidelines and requirements put in place by authorities including the MHRA, EMA and FDA.

# Deep Learning



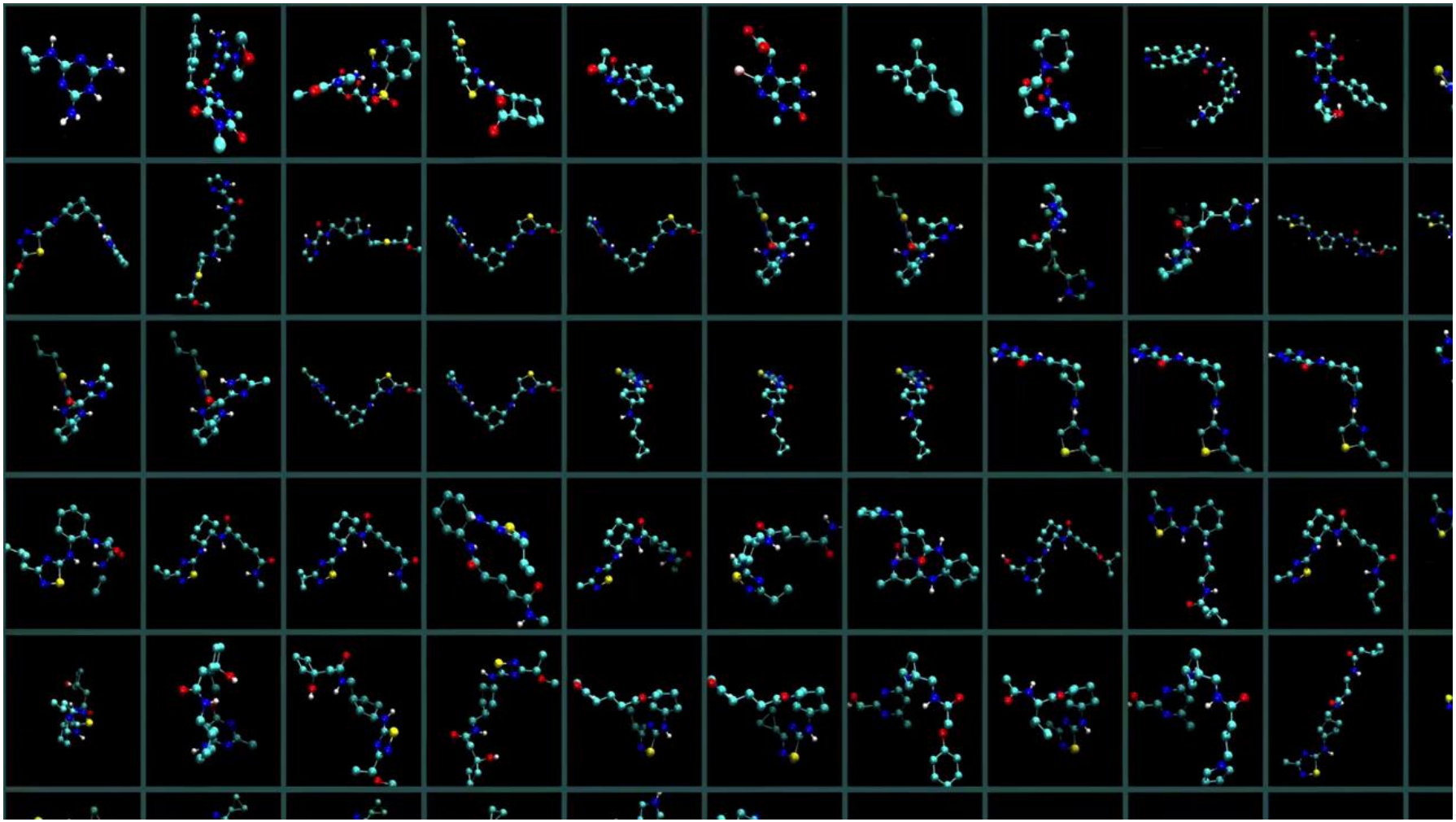
# AI-based Hit to Lead



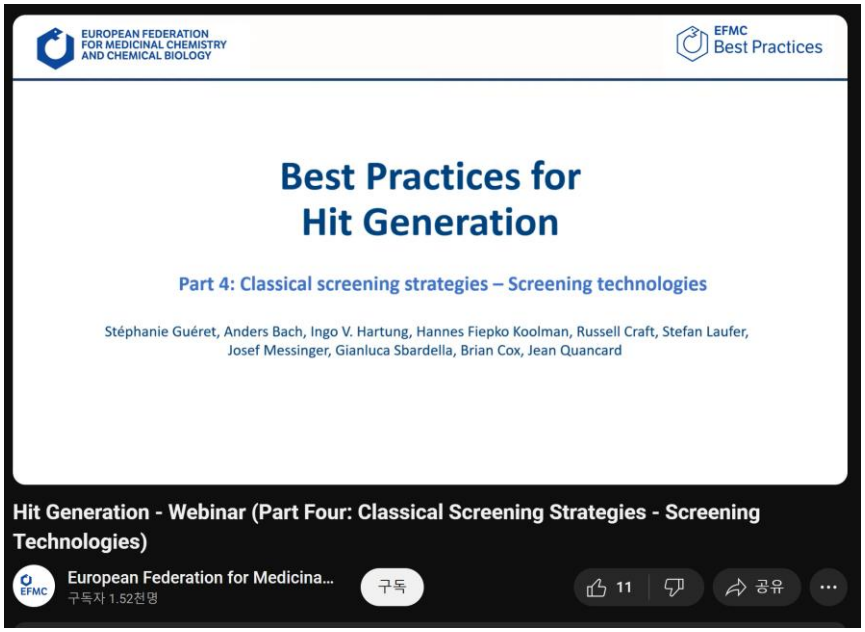

[그림 11] 인공지능을 활용한 신약후보물질 탐색

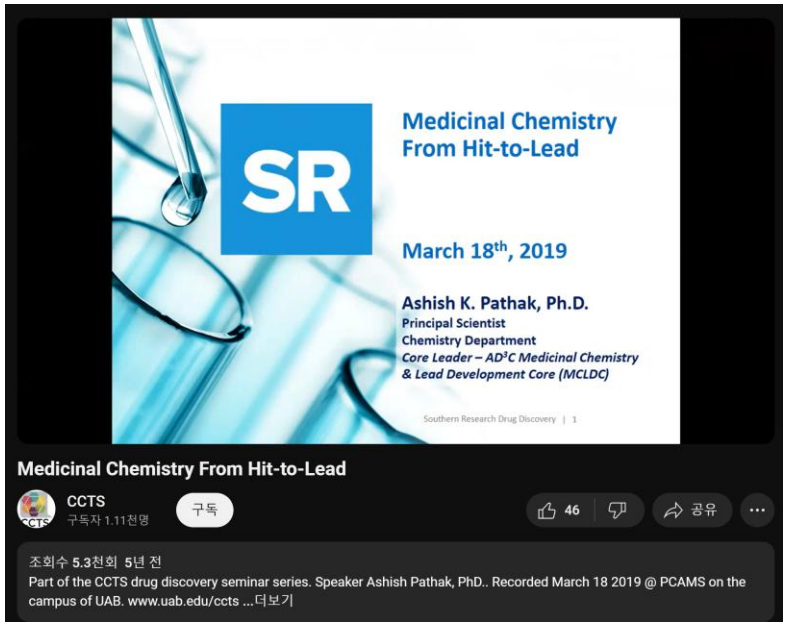
| 1단계 | 2단계 | 3단계 | 4단계 |
|---|---|---|---|
| Database | AI | Prioritization | Synthesis/Evaluation |
| · 제약사 요구 사항에 따른 내용 조사<br>· 관련 개발 동향 조사<br>· 학습데이터 선정 | · 실험 목적에 따른 인공지능 알고리즘 구현 및 데이터 정제 진행<br>· AI 학습<br>· 딥러닝 튜닝 | · Drug-likeness properties 값을 기준으로 필터<br>· 물질 유사도 검토<br>· 특허 신규성 검토<br>· In silico docking study 진행 | · 선별 구조에 대한 합성법 확립<br>· 물질 합성<br>· in vitro 물질 평가<br>· in vivo 물질 효력 및 독성 평가<br>· 약리기전 연구 |

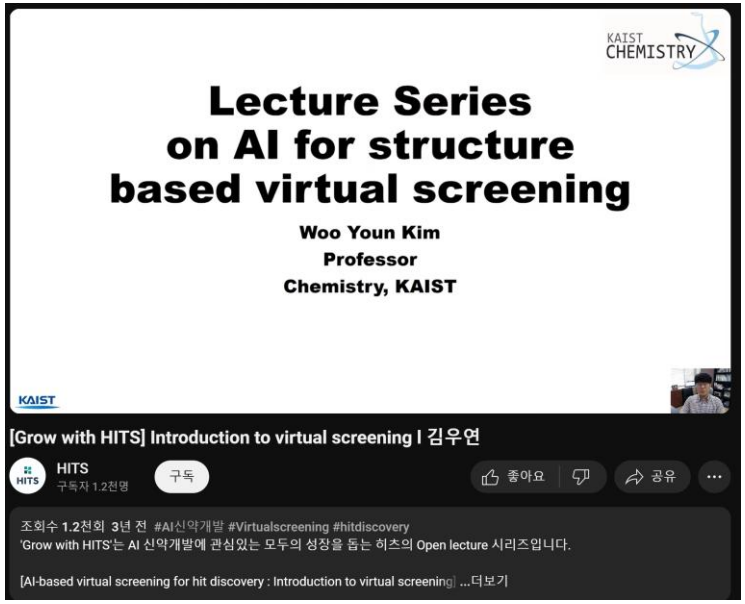
[그림 12] 인공지능 신약후보물질 개발을 위한 프로토콜

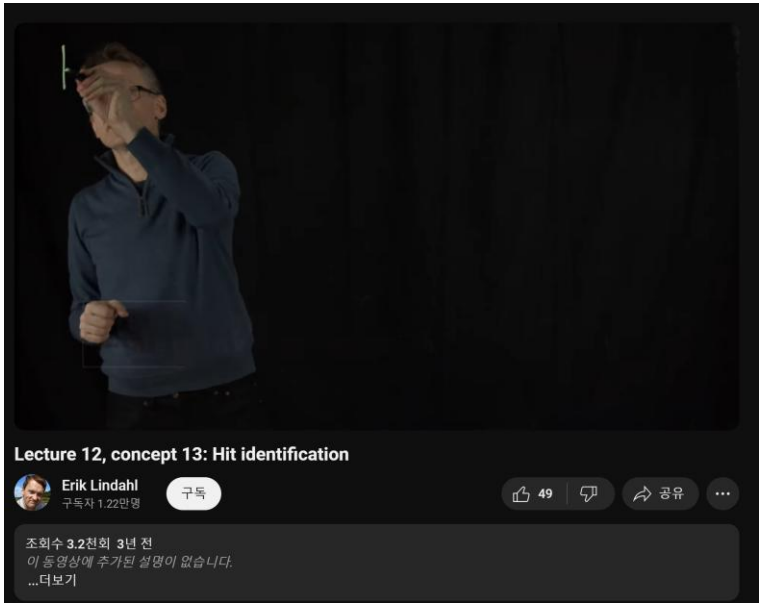


# Youtube E-learning



# Questions?